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B.Tech. Degree VI Semester Regular/Supplementary Examination May 2024

CE 19-201-0606 (IE) SUSTAINABLE CONSTRUCTION TECHNIQUES
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Identify and understand the concept of sustainability in construction practices.
 CO2: Compare the buildings based on the concept of green building rating systems.
 CO3: Identify and understand the usability of locally available materials in building construction.
 CO4: Understand various innovative techniques of cost reductions for superstructure.
 CO5: Understand various innovative techniques on alternative roofing.
 CO6: Gain knowledge on various mud construction techniques.
 CO7: Identify and understand the usability of waste materials in building construction.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate,
 L6 – Create

PO – Programme Outcome

PART A(Answer *ALL* questions)

(8 × 3 = 24)

		Marks	BL	CO	PO
I.	(a) Illustrate Zero Energy Building.	3	L3	1	1,3,7,12
	(b) Enumerate the Dimensions of Sustainability.	3	L5	1	1,6,7,8,10
	(c) List out the planning aspects to be considered in reducing the cost of the construction.	3	L1	4	1,2,3
	(d) Explain Sand Piles.	3	L2	4	1,6,7,8,10
	(e) Why is Precast Concrete more Sustainable?	3	L2	3	1,6,7,8,10
	(f) Enumerate the purpose of Filler Slab.	3	L1	4	1,3
	(g) List out the application of Ferro cement.	3	L1	6	1,2,3
	(h) Illustrate the Wattle and Daub method of mud construction.	3	L3	7	1,2,3

PART B

(4 × 12 = 48)

II.	Discuss the Green Rating Systems in India.	12	L5	2	1,2,7
OR					
III.	Suggest any two examples of sustainable construction practices for replacing conventional alternatives. Justify your answer.	12	L3	1	1,3,7,12
IV.	Explain the application of locally available materials in sustainable construction.	12	L5	3	1,6,7,8,10
OR					
V.	Explain the following types of cost effective foundations with neat figures:	12	L4	4	1,2,3
	(i) Brick Arch Foundation				
	(ii) Stub Foundation.				

(P.T.O.)

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		Marks	BL	CO	PO
VI.	Illustrate the following roofing techniques: (i) Filler Slabs (ii) Precast reinforced concrete channel units (iii) Funicular Shells.	12	L2	5	1,2,3
	OR				
VII.	Write short notes on the following: (i) Rat trap bond masonry (ii) Stabilized soil blocks (iii) Hollow Concrete blocks.	12	L2	4	1,2,3
VIII.	Explain the various Field tests and Lab tests for identification of suitable Soil for Mud construction.	12	L1	6	1,2,3
	OR				
IX.	What is Prefabrication? Explain the benefits and challenges of Prefabrication.	12	L5	7	1,3,7

Blooms's Taxonomy Level's

L1 – 22.8%, L2 – 20.6%, L3 – 13%, L4 – 8%, L5 – 35.6%.
